5.14.7 *Lattice structure of IIR system*

Let us consider an all-pole system with system function

$$H(z) = \frac{1}{1 + \sum_{k=1}^{N} a_N z^{-k}} = \frac{1}{A_N(z)} \tag{5.127}$$

The difference equation for this IIR system is

$$y(n) = -\sum_{k=1}^{N} a_N(k) y(n-k) + x(n) \tag{5.128}$$

or

$$x(n) = y(n) + \sum_{k=1}^{N} a_N(k) y(n-k) \tag{5.129}$$

For $N = 1$

$$x(n) = y(n) + a_1(1) y(n-1) \tag{5.130}$$

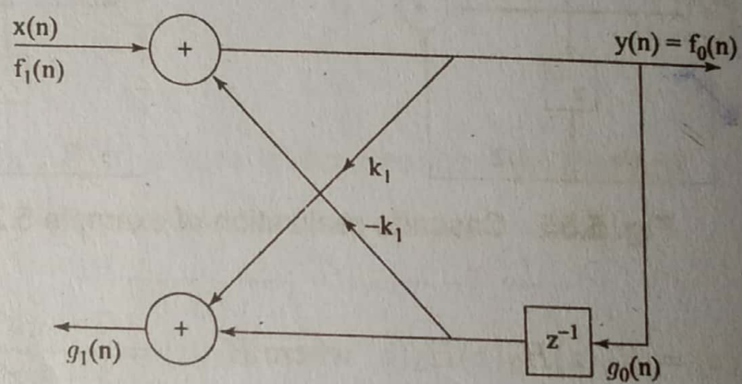

Fig. 5.57 Single stage all-pole lattice filter

The Eq. (5.130) can be realised in lattice structure as shown in Fig. 5.57 from which we can obtain

$$x(n) = f_1(n) \tag{5.131}$$

$$y(n) = f_0(n) = f_1(n) - k_1 g_0(n-1)$$

$$= x(n) - k_1 y(n-1) \tag{5.132a}$$

$$\Rightarrow \quad x(n) = y(n) + k_1 y(n-1) \tag{5.132b}$$

$$g_1(n) = k_1 f_0(n) + g_0(n-1)$$

$$= k_1 y(n) + y(n-1) \tag{5.133}$$

Comparing Eq. (5.130) and Eq. (5.132b) we have

$$k_1 = a_1(1) \tag{5.134}$$

Now, let us consider for the case $N = 2$, then

$$x(n) = f_2(n) \tag{5.135a}$$

$$y(n) = x(n) - a_2(1)y(n-1) - a_2(2)y(n-2) \tag{5.135b}$$

This output can also be obtained from a two-stage lattice filter as shown in Fig. 5.58 from which we have

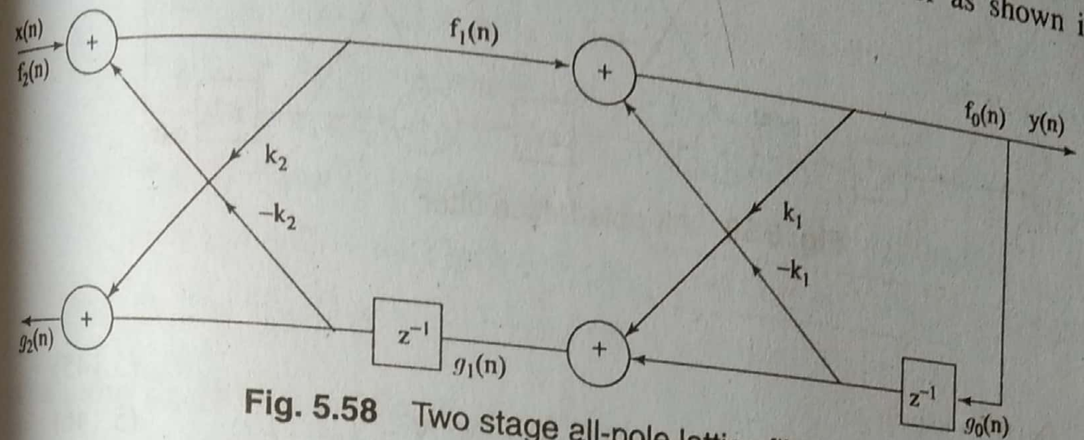

Fig. 5.58 Two stage all-pole lattice filter.

$$f_2(n) = x(n) \tag{5.136}$$

$$f_1(n) = f_2(n) - k_2 g_1(n-1) \tag{5.137}$$

$$g_2(n) = k_2 f_1(n) + g_1(n-1) \tag{5.138}$$

$$f_0(n) = f_1(n) - k_1 g_0(n-1) \tag{5.139}$$

$$g_1(n) = k_1 f_0(n) + g_0(n-1) \tag{5.140}$$

$$\begin{aligned} y(n) &= f_0(n) = g_0(n) \qquad (5.141)\\ &= f_1(n) - k_1 g_0(n-1)\\ &= f_2(n) - k_2 g_1(n-1) - k_1 g_0(n-1)\\ &= f_2(n) - k_2[k_1 f_0(n-1) + g_0(n-2)] - k_1 g_0(n-1) \end{aligned}$$

Using Eq. (5.141)

$$\begin{aligned} y(n) &= f_2(n) - k_2[k_1 y(n-1) + y(n-2)] - k_1 y(n-1)\\ &= x(n) - k_1(1+k_2)y(n-1) - k_2 y(n-2) \qquad (5.142) \end{aligned}$$

Similarly

$$g_2(n) = k_2 y(n) + k_1(1+k_2)y(n-1) + y(n-2) \tag{5.143}$$

Comparing Eq. (5.135b) and Eq. (5.142) we get

$$a_2(0) = 1;\, a_2(1) = k_1(1+k_2);\, a_2(2) = k_2 \tag{5.144}$$

For a N-stage IIR filter realized in lattice structure as shown in Fig. 5.59 we get

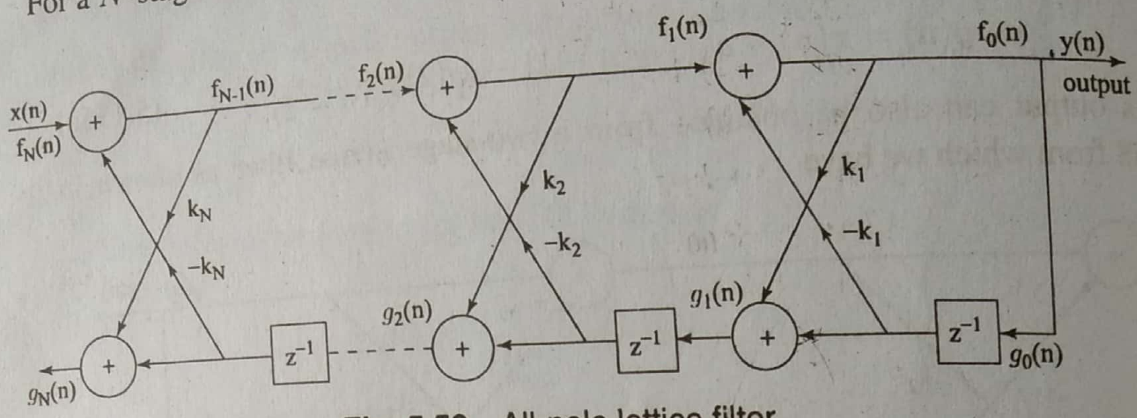

Fig. 5.59 All-pole lattice filter

$$f_N(n) = x(n) \tag{5.145}$$
$$f_{m-1}(n) = f_m(n) - k_m g_{m-1}(n-1) \quad m = N, N-1, \ldots 1 \tag{5.146}$$
$$g_m(n) = k_m f_{m-1}(n) + g_{m-1}(n-1) \quad m = N, N-1, \ldots 1 \tag{5.147}$$
$$y(n) = f_0(n) = g_0(n) \tag{5.148}$$

5.14.8 *Conversion from Lattice Structure to direct form*

For $N = 3$ we can write the difference equation for an IIR filter as

$$y(n) = x(n) - a_3(1)y(n-1) - a_3(2)y(n-2) - a_3(3)y(n-3) \tag{5.149}$$

and from the lattice structure, we can write

$$y(n) = f_0(n) = g_0(n) \tag{5.150}$$
$$x(n) = f_3(n) = f_2(n) + k_3 g_2(n-1) \tag{5.151}$$
$$g_3(n) = k_3 f_2(n) + g_2(n-1) \tag{5.152}$$

Substituting $m = 2$ in Eq. (5.146) and in Eq. (5.147) we get

$$f_2(n) = f_1(n) + k_2 g_1(n-1) \tag{5.153}$$

and

$$g_2(n-1) = k_2 f_1(n-1) + g_1(n-2) \tag{5.154}$$

Substitute Eq. (5.153) and Eq. (5.154) in Eq. (5.151) we obtain

$$\begin{aligned}
x(n) &= f_2(n) + k_3 g_2(n-1) \\
&= f_1(n) + k_2 g_1(n-1) + k_3[k_2 f_1(n-1) + g_1(n-2)] \\
&= f_0(n) + k_1 g_0(n-1) + k_2[k_1 f_0(n-1) + g_0(n-2)] \\
&\quad + k_3 k_2[f_0(n-1) + k_1 g_0(n-2)] \\
&\quad + k_3[k_1 f_0(n-2) + g_0(n-3)] \qquad (5.155) \\
&= y(n) + [k_1(1+k_2) + k_2 k_3] y(n-1) \\
&\quad + [k_2 + k_1 k_3(1+k_2)] y(n-2) + k_3 y(n-3) \qquad (5.156) \\
&= y(n) + [a_2(1) + a_2(2) a_3(3)] y(n-1) + [a_2(2) + a_2(1) a_3(3)] \\
&\quad \times y(n-2) + a_3(3) y(n-3) \quad \boxed{\because a_2(1) = k_1(1+k_2)} \qquad (5.157) \\
&= y(n) + a_3(1) y(n-1) + a_3(2) y(n-2) + a_3(3) y(n-3)
\end{aligned}$$

Comparing Eq. (5.149) and Eq. (5.157) we get

$$\begin{aligned}
a_3(0) = 1; a_3(1) &= a_2(1) + a_2(2) a_3(3) \\
a_3(2) &= a_2(2) + a_2(1) a_3(3) \\
k_3 &= a_3(3) \qquad (5.158)
\end{aligned}$$

In general $a_m(0) = 1; a_m(k) = a_{m-1}(k) + a_m(m) a_{m-1}(m-k)$

$$a_m(m) = k_m \qquad (5.159)$$

The Eq. (5.159) can be used to convert lattice structure to direct form.

5.14.9 *Conversion from direct form to Lattice structure*

For a 3-stage IIR system

$$x(n) = f_3(n) = f_2(n) + k_3 g_2(n-1) \qquad (5.160)$$

$$g_3(n) = k_3 f_2(n) + g_2(n-1) \qquad (5.161)$$

from which

$$g_2(n-1) = g_3(n) - k_3 f_2(n) \qquad (5.162)$$

Substitute Eq. (5.162) in Eq. (5.160) we get

$$f_3(n) = f_2(n) + k_3[g_3(n) - k_3 f_2(n)]$$

and

$$f_2(n) = \frac{f_3(n) - k_3 g_3(n)}{1 - k_3^2}$$

$$= \frac{\begin{array}{l} y(n) + a_3(1)y(n-1) + a_3(2)y(n-2) + a_3(3)y(n-3) \\ -k_3 a_3(3)y(n) - k_3 a_3(2)y(n-2) \\ -k_3 a_3(n)y(n-1) - a_3 y(n-3) \end{array}}{1 - k_3^2}$$

$$= y(n) + \frac{a_3(1) - a_3(3)a_3(2)}{1 - a_3^2(3)} y(n-1) + \frac{a_3(2) - a_3(2)a_3(3)}{1 - a_3^2(3)} y(n-2) \tag{5.163}$$

Comparing Eq. (5.163) and Eq. (5.135b) we have

$$a_2(0) = 1; a_2(1) = \frac{a_3(1) - a_3(3)a_3(2)}{1 - a_3^2(3)}; a_2(2) = \frac{a_3(2) - a_3(2)a_3(3)}{1 - a_3^2(3)} \tag{5.164}$$

In general $a_{m-1}(0) = 1$

$$k_m = a_m(m)$$

$$a_{m-1}(k) = \frac{a_m(k) - a_m(m)a_m(m-k)}{1 - a_m^2(m)} \tag{5.165}$$

The Eq. (5.165) can be used to convert direct form to lattice structure.

5.14.10 *Lattice-Ladder Structure*

A general IIR filter containing both poles and zeros can be realized using an all pole lattice as the basic building block.

Consider an IIR filter with system function

$$H(z) = \frac{B_M(z)}{A_N(z)} = \frac{\sum_{k=0}^{N} b_M(k)z^{-k}}{1 + \sum_{k=1}^{N} a_N(k)z^{-k}} \tag{5.166}$$

where $N \geq M$. A lattice structure for Eq. (5.166) can be constructed by first realizing an all-pole lattice co-efficients $k_m, 1 \leq m \leq N$ for the denominator $A_N(z)$, and then adding a ladder part as shown in Fig. 5.60 for $M = N$. The output of the ladder part can be expressed as a weighted linear combination of $\{g_m(n)\}$.

Now the output is given by

$$y(n) = \sum_{m=0}^{M} c_m g_m(n) \tag{5.167}$$

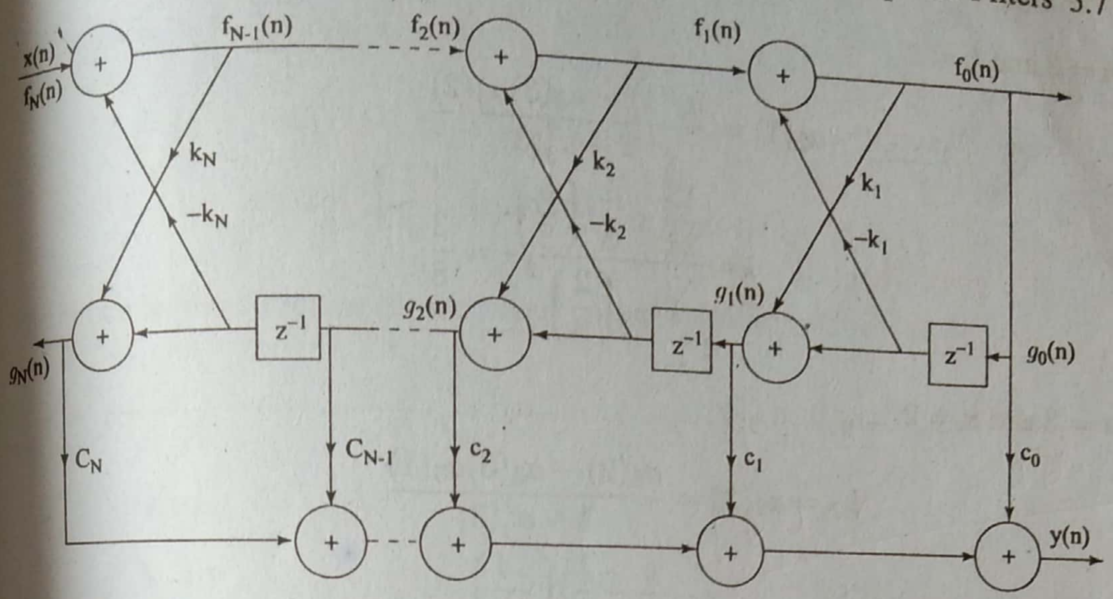

Fig. 5.60 Lattice-ladder structure for realizing a pole-zero IIR filter.

where $\{c_m\}$ are called the ladder co-efficients and can be obtained using the recursive relation.

$$c_m = b_m - \sum_{i=m+1}^{M} c_i a_i(i-m); m = M, M-1, \dots 0 \tag{5.168}$$

Example 5.29 Convert the following pole zero IIR filter into a lattice-ladder structure.

$$H(z) = \frac{1+2z^{-1}+2z^{-2}+z^{-3}}{1+\frac{13}{24}z^{-1}+\frac{5}{8}z^{-2}+\frac{1}{3}z^{-3}}$$

9 4 5 3 0 7 4 7

Solution

Given $b_M(z) = 1+2z^{-1}+2z^{-2}+z^{-3}$

$$A_N(z) = 1+\frac{13}{24}z^{-1}+\frac{5}{8}z^{-2}+\frac{1}{3}z^{-3}$$

$$a_3(0) = 1; a_3(1) = \frac{13}{24}; a_3(2) = \frac{5}{8}; a_3(3) = \frac{1}{3}$$

$$k_3 = a_3(3) = \frac{1}{3}$$

From Eq. (5.165) we have

$$a_{m-1}(k) = \frac{a_m(k) - a_m(m)a_m(m-k)}{1-a_m^2(m)}$$

For $m = 3$ and $k = 1$

$$a_2(1) = \frac{a_3(1) - a_3(3)a_3(2)}{1 - a_3^2(3)}$$

$$= \frac{\frac{13}{24} - \frac{1}{3}\left(\frac{5}{8}\right)}{1 - \left(\frac{2}{3}\right)^2} = \frac{3}{8}$$

For $m = 3$ and $k = 2$

$$k_2 = a_2(2) = \frac{a_3(2) - a_3(3)a_3(1)}{1 - a_3^2(3)}$$

$$= \frac{\frac{5}{8} - \frac{1}{3}\left(\frac{13}{24}\right)}{1 - \left(\frac{1}{3}\right)^2} = \frac{1}{2}$$

For $m = 2$ and $k = 1$

$$k_1 = a_1(1) = \frac{a_2(1) - a_2(2)a_2(1)}{1 - a_2^2(2)}$$

$$= \frac{\frac{3}{8} - \frac{1}{2}\left(\frac{3}{8}\right)}{1 - \left(\frac{1}{2}\right)^2} = \frac{1}{4}$$

Therefore, for lattice structure

$$k_1 = \frac{1}{4}; \quad k_2 = \frac{1}{2}; \quad k_3 = \frac{1}{3}$$

For ladder structure

$$c_m = b_m - \sum_{i=m+1}^{M} c_i a_i(i - m); \quad m = M, M-1, \ldots 0$$

$$c_3 = b_3 = 1$$

$$c_2 = b_2 - c_3 a_3(1)$$

$$= 2 - 1\left(\frac{13}{24}\right) = 1.4583$$

$$c_1 = b_1 - \sum_{i=2}^{3} c_i a_i(i - m)$$

$$= b_1 - [c_2 a_2(1) + c_3 a_3(2)]$$

$$= 2 - [(1.4583)(3/8) + 5/8] = 0.8281$$

$$c_0 = b_0 - \sum_{i=1}^{3} c_i a_i(i-m)$$

$$= b_0 - [c_1 a_1(1) + c_2 a_2(2) + c_3 a_3(3)]$$

$$= 1 - \left[0.8281\left(\frac{1}{4}\right) + 1.4583\left(\frac{1}{2}\right) + \frac{1}{3}\right] = -0.2695.$$

The lattice ladder structure for the given pole-zero filter is shown in Fig. 5.61.

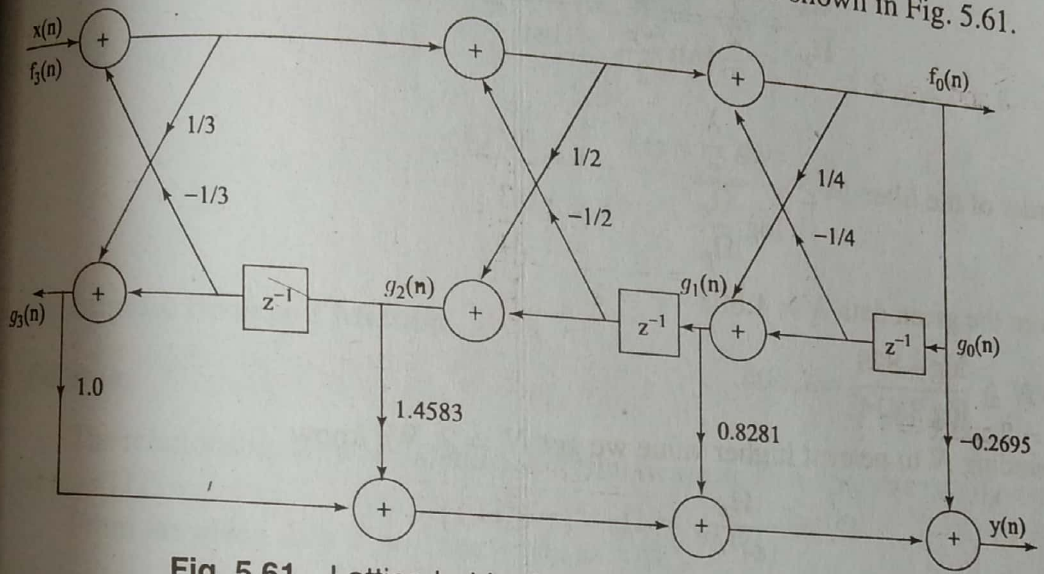

Fig. 5.61 Lattice-ladder form for the example 5.29.

To convert a lattice-ladder form into a direct form, we first use Eq. (5.159) to determine $\{a_N(k)\}$ and then use Eq. (5.168) recursively to obtain $b_M(k)$.

Practice Problem 5.16 Convert the following pole-zero IIR filter into a lattice-ladder structure

$$H(z) = \frac{1 + z^{-1} + z^{-2}}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}.$$

Additional Examples

Example 5.30 Design a digital Butterworth filter satisfying the constraints

$$0.707 \le |H(e^{j\omega})| \le 1 \quad \text{for } 0 \le \omega \le \frac{\pi}{2}$$

$$|H(e^{j\omega})| \le 0.2 \quad \text{for } \frac{3\pi}{4} \le \omega \le \pi$$

with $T = 1$ sec using (a) The bilinear transformation (b) Impulse invariance. Realize the filter in each case using the most convenient realization form.

(AU ECE May'07) (AU ECE Nov'06)

Solution

(a) Bilinear transformation

Given data $\dfrac{1}{\sqrt{1+\varepsilon^2}} = 0.707;\quad \dfrac{1}{\sqrt{1+\lambda^2}} = 0.2; \omega_p = \dfrac{\pi}{2}; \omega_s = \dfrac{3\pi}{4}.$

The analog frequency ratio is

$$\frac{\Omega_s}{\Omega_p} = \frac{\frac{2}{T}\tan\frac{\omega_s}{2}}{\frac{2}{T}\tan\frac{\omega_p}{2}} = \frac{\tan\frac{3\pi}{8}}{\tan\frac{\pi}{4}} = 2.414$$

The order of the filter $N \geq \dfrac{\log\frac{\lambda}{\varepsilon}}{\log\frac{\Omega_s}{\Omega_p}}$.

From the given data $\lambda = 4.898 \quad \varepsilon = 1.$

So $N \geq \dfrac{\log 4.898}{\log 2.414} = 1.803.$

Rounding N to nearest higher value we get $N = 2$. We know

$$\Omega_c = \frac{\Omega_p}{(\varepsilon)^{1/N}} = \Omega_p \quad (\because \varepsilon = 1)$$

$$= \frac{2}{T}\tan\frac{\omega_p}{2} = 2\tan\frac{\pi}{4} = 2\,\text{rad/sec}$$

The transfer function of second order normalized Butterworth filter is

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

$H_a(s)$ for $\Omega_c = 2$ rad/sec can be obtained by substituting $s \to s/2$ in $H(s)$

$$\text{i.e.,}\quad H_a(s) = \frac{1}{(s/2)^2 + \sqrt{2}\cdot(s/2) + 1}$$

$$= \frac{4}{s^2 + 2.828s + 4}$$

By using bilinear transformation $H(z)$ can be obtained as

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$\text{Thus } H(z) = \frac{4}{s^2 + 2.828s + 4}\Bigg|_{s=2\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} \qquad (\because T = 1\,\text{sec})$$

$$= \frac{4(1+z^{-1})^2}{4(1-z^{-1})^2 + 5.656(1-z^{-2}) + 4(1+z^{-1})^2}$$

$$= \frac{0.2929(1+z^{-1})^2}{1 + 0.1716z^{-2}}$$

The above system function can be realized in direct form II as shown in Fig. 5.62.

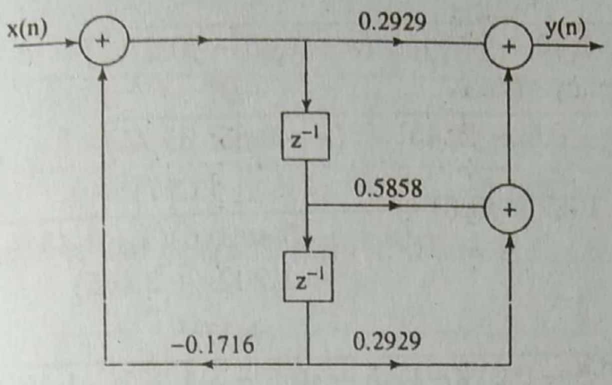

Fig. 5.62

(b) Impulse Invariant Method

Solution

The relationship between analog & digital frequencies in Impulse invariant method is $\omega = \Omega T$.

From the given data $T = 1$ sec i.e., $\omega = \Omega$

$$\Rightarrow \quad \Omega_p = \omega_p; \quad \Omega_s = \omega_s$$

We know $\lambda = 4.898$; $\varepsilon = 1$.

The order of the filter

$$N \geq \frac{\log \frac{\lambda}{\varepsilon}}{\log \frac{\Omega_s}{\Omega_p}} = \frac{\log 4.989}{\log \frac{3\pi/4}{\pi/2}}$$

$$N \geq 3.924$$

i.e., $N = 4$

The transfer function of a fourth order normalized Butterworth filter is

$$H(s) = \frac{1}{(s^2 + 0.76537 + 1)(s^2 + 1.8477s + 1)}$$

As $\varepsilon = 1$: $\quad \Omega_p = \Omega_c = 0.5\pi = 1.57$

$$H_a(s) = H(s)\Big|_{s \to \frac{s}{1.57}}$$

$$= \frac{(1.57)^4}{(s^2 + 1.202s + 2.465)(s^2 + 2.902s + 2.465)}$$

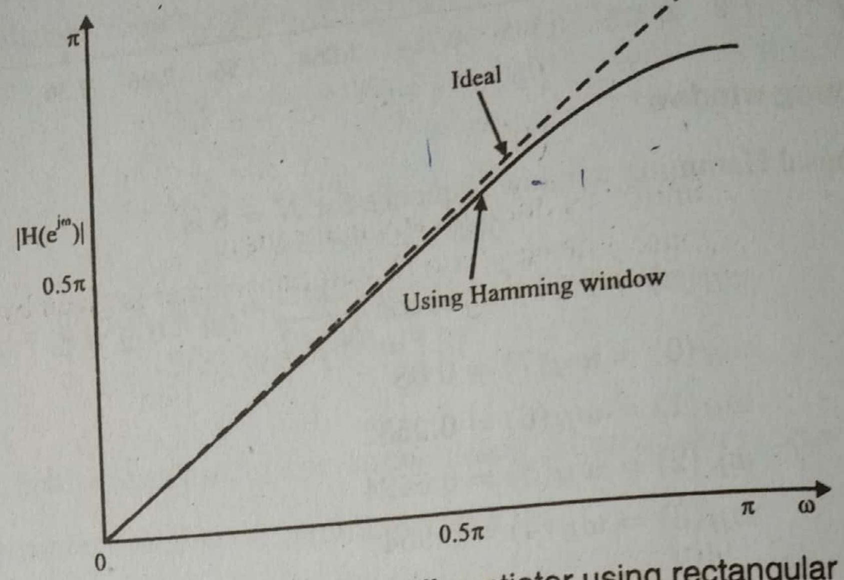

Fig. 6.52 Frequency response of differentiator using rectangular window

6.8 Hilbert Transformers

The frequency response of an ideal Hilbert transformer is given by

$$H_d(e^{j\omega}) = \begin{cases} -j & \text{for } 0 < \omega < \pi \\ j & \text{for } -\pi < \omega < 0 \end{cases}$$

The magnitude response of a Hilbert transformer is 1 for all frequencies; thus it is an all-pass filter. But it introduces a phase shift of $-90°$ for $\omega > 0$ and $90°$ for $\omega < 0$. Hilbert transformers are used in communication systems, particularly in the generation of single-sideband modulated signals, radar signal processing, and speech signal processing.

The impulse response of an ideal Hilbert transformer is

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(e^{j\omega})e^{j\omega n}\,d\omega$$

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{0} j\,e^{j\omega n}\,d\omega - \int_{0}^{\pi} j\,e^{j\omega n}\,d\omega\right]$$

$$= \frac{1}{2\pi}\left[j\,\frac{e^{j\omega n}}{jn}\bigg|_{-\pi}^{0} - j\,\frac{e^{j\omega n}}{jn}\bigg|_{0}^{\pi}\right]$$

$$= \frac{1}{2\pi n}\left[1 - e^{-j\pi n} - e^{j\pi n} + 1\right]$$

$$= \frac{1}{2\pi n}\left[2 - (e^{j\pi n} + e^{-j\pi n})\right]$$

$$= \frac{1}{2\pi n}[2 - 2\cos \pi n] = \frac{1 - \cos \pi n}{\pi n}$$

$$= \frac{2\sin^2\left(\frac{\pi n}{2}\right)}{\pi n} \quad \text{for } n \neq 0$$
$$= 0 \quad \text{for } n = 0$$

$$h_d(n) = \begin{cases} 2\sin^2\left(\frac{\pi n}{2}\right) & \text{for } n \neq 0 \\ 0 & \text{for } n = 0 \end{cases}$$

Note that $h_d(n)$ is infinite in duration and non-causal.

The frequency response a linear phase Hilbert transformer is given by

$$H_d(e^{j\omega}) = -j\,e^{-j\alpha\omega} \quad \text{for } 0 < \omega < \pi$$
$$= j\,e^{-j\alpha\omega} \quad \text{for } -\pi < \omega < 0$$

where $$\alpha = \frac{N-1}{2}$$

The linear phase introduces a delay of α samples.

The impulse response of a linear phase Hilbert transformer is

$$h_d(n) = \frac{2\sin^2\frac{(n-\alpha)\pi}{2}}{\pi(n-\alpha)}$$

For N odd, the truncated $h_d(n)$ is a type III FIR filter. We know that for a type III FIR filter $\overline{H}(e^{j\omega})$ is zero at $\omega = 0$ and $\omega = \pi$. Then it is impossible to design an all-pass Hilbert transformer. But in practical signal processing applications an all pass Hilbert transformer is unnecessary.

Example 6.14 Design an ideal Hilbert transformer having frequency response

$$H(e^{j\omega}) = j \quad \text{for} \quad -\pi \leq \omega \leq 0$$
$$= -j \quad \text{for} \quad 0 \leq \omega \leq \pi$$

Using (a) rectangular window (b) Blackman window

For $N = 11$ plot the frequency response in both cases. (AU ECE May'07)

Solution

The ideal frequency response is shown in Fig. 6.53.

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(e^{j\omega})e^{j\omega n}d\omega$$
$$= \frac{1}{2\pi}\left[\int_{-\pi}^{0} je^{j\omega n}d\omega + \int_{0}^{\pi} -je^{j\omega n}d\omega\right]$$
$$= \frac{1-\cos\pi n}{\pi n}$$

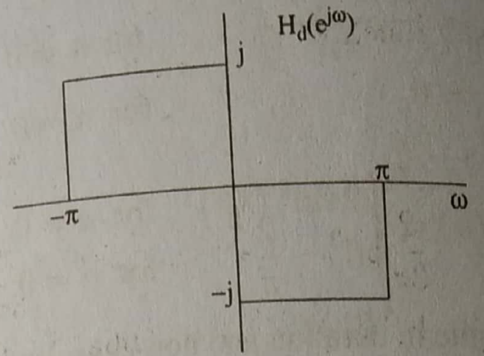

Fig. 6.53 Frequency response of ideal Hilbert transformer

The filter coefficients are antisymmetrical about $n = 0$, satisfying $h_d(n) = -h_d(-n)$. For $n = 0$

$$h_d(0) = \frac{j}{2\pi}\left[\int_{-\pi}^{0} d\omega + \int_{0}^{\pi} d\omega\right]$$

$$= \frac{j}{2\pi}(0 + \pi - \pi - 0) = 0$$

$$h_d(1) = -h_d(-1) = \frac{1-\cos\pi}{\pi} = \frac{2}{\pi}$$

$$h_d(2) = -h_d(-2) = \frac{1-\cos 2\pi}{2\pi} = 0$$

$$h_d(3) = -h_d(-3) = \frac{1-\cos 3\pi}{3\pi} = \frac{2}{3\pi}$$

$$h_d(4) = -h_d(-4) = \frac{1-\cos 4\pi}{4\pi} = 0$$

$$h_d(5) = -h_d(-5) = \frac{1-\cos 5\pi}{5\pi} = \frac{2}{5\pi}$$

(a) **Rectangular window**

$$h(n) = h_d(n)w_R(n) = h_d(n) \quad \text{for} \quad -5 \le n \le 5$$

$$h(1) = -h(-1) = \frac{2}{\pi}$$

$$h(2) = -h(-2) = 0$$

$$h(3) = -h(-3) = \frac{2}{3\pi}$$

$$h(4) = -h(-4) = 0$$

$$h(5) = -h(-5) = \frac{2}{5\pi}$$

The transfer function of the Hilbert-transformer is

$$H(z) = \frac{2}{\pi}(z - z^{-1}) + \frac{2}{3\pi}(z^3 - z^{-3}) + \frac{2}{5\pi}(z^5 - z^{-5})$$

The realizable transfer function

$$
\begin{aligned}
H'(z) &= z^{-5}H(z) \\
&= \frac{2}{5\pi} + \frac{2}{3\pi}z^{-2} + \frac{2}{\pi}z^{-4} - \frac{2}{\pi}z^{-6} - \frac{2}{3\pi}z^{-8} - \frac{2}{5\pi}z^{-10} \\
&= 0.127 + 0.2122z^{-2} + 0.6366z^{-4} - 0.6366z^{-6} \\
&\quad - 0.2122z^{-8} - 0.127z^{-10}
\end{aligned}
$$

The causal sequence of Hilbert transformer is

$$
\begin{aligned}
h(0) &= -h(10) = 0.1273 \\
h(1) &= h(9) = h(3) = h(7) = h(5) = 0 \\
h(2) &= -h(8) = 0.2122 \\
h(4) &= -h(6) = 0.6366
\end{aligned}
$$

$$\overline{H}(e^{j\omega}) = \sum_{n=1}^{5} c(n)\sin\omega n$$

$$
\begin{aligned}
c(n) &= 2h\left(\frac{N-1}{2} - n\right) \\
c(1) &= 2h(4) = 1.2732 \\
c(2) &= 2h(3) = 0 \\
c(3) &= 2h(2) = 0.4244 \\
c(4) &= 2h(1) = 0 \\
c(5) &= 2h(0) = 0.2546
\end{aligned}
$$

$$\overline{H}(e^{j\omega}) = 1.2732\sin\omega + 0.4244\sin 3\omega + 0.2546\sin 5\omega$$

$$H(e^{j\omega}) = -j\overline{H}(e^{j\omega})$$

$$H(e^{j\omega}) = -j[1.2732\sin\omega + 0.4244\sin 3\omega + 0.2546\sin 5\omega]$$

ω	0	10	20	30	45	60	75
$H(e^{j\omega})$	0	$-j0.627$	$-j1.05$	$-j1.19$	$-j1.02$	$-j0.88$	$-j0.99$

90	105	120	135	150	165	180
$-j1.1$	$-j0.996$	$-j0.88$	$-j1.02$	$-j1.188$	$-j0.8755$	0

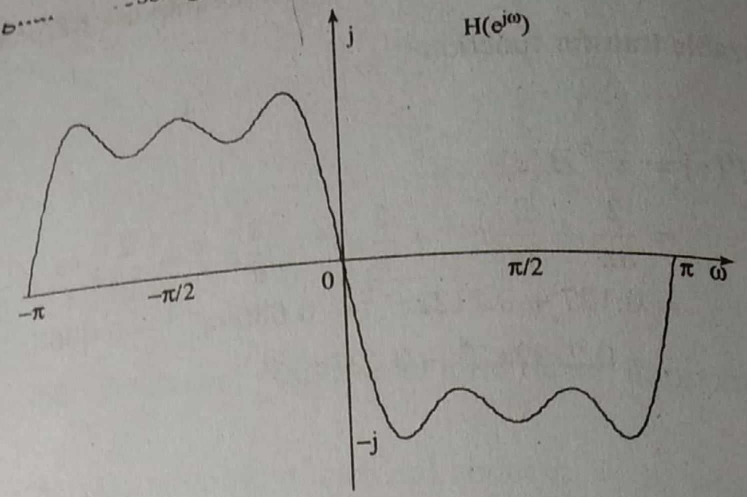

Fig. 6.54 Frequency response of Hilbert transformer using rectangular window

(b) Blackman window

The Blackman window sequence for $N = 11$ is

$$w_B(n) = 0.42 + 0.5\cos\frac{\pi n}{5} + 0.08\cos\frac{2\pi n}{5} \quad \text{for} \quad -5 \le n \le 5$$
$$= \quad 0 \quad \text{otherwise}$$

$$w_B(0) = 1$$
$$w_B(1) = w_B(-1) = 0.849$$
$$w_B(2) = w_B(-2) = 0.509$$
$$w_B(3) = w_B(-3) = 0.2$$
$$w_B(4) = w_B(-4) = 0.04$$
$$w_B(5) = w_B(-5) = 0$$

The coefficients of Hilbert transformer are

$$h(n) = h_d(n)w_B(n) \quad \text{for} \quad -5 \le n \le 5$$
$$= \quad 0 \quad \text{otherwise}$$
$$h(0) = h_d(0)w_B(0) = (0)(1) = 0$$
$$h(1) = -h(-1) = h_d(1)w_B(1) = \left(\frac{2}{\pi}\right)(0.849) = 0.5405$$
$$h(2) = -h(-2) = h_d(2)w_B(2) = (0)(0.509) = 0$$
$$h(3) = -h(-3) = h_d(3)w_B(3) = \left(\frac{2}{3\pi}\right)(0.2) = 0.0423$$
$$h(4) = -h(-4) = h_d(4)w_B(4) = (0)(0.04) = 0$$
$$h(5) = -h(-5) = h_d(5)w_B(5) = \left(\frac{2}{5\pi}\right)(0) = 0$$

The realizable transfer function is

$$
\begin{aligned}
H'(z) &= z^{-5}H(z)\\
&= z^{-5}[0.54(z - z^{-1}) + 0.0424(z^3 - z^{-3})]\\
&= 0.0424z^{-2} + 0.54z^{-4} - 0.54z^{-6} - 0.042z^{-8}
\end{aligned}
$$

The causal coefficients of Hilbert transformer are

$$
\begin{aligned}
h(0) &= h(10) = h(1) = h(9) = h(3) = h(7) = h(5) = 0\\
h(2) &= -h(8) = 0.424\\
h(4) &= -h(6) = 0.54\\
\overline{H}(e^{j\omega}) &= \sum_{n=1}^{5} c(n)\sin n\omega\\
&= 1.08\sin\omega + 0.089\sin 3\omega\\
H(e^{j\omega}) &= -j\overline{H}(e^{j\omega})
\end{aligned}
$$

ω	0	15	30	45	60	75	90	105	120	135	150	165	180
$H(e^{j\omega})$	0	$-j0.339$	$-j0.624$	$-j0.823$	$-j0.935$	$-j0.984$	$-j0.996$	$-j0.984$	$-j0.935$	$-j0.823$	$-j0.624$	$-j0.339$	0

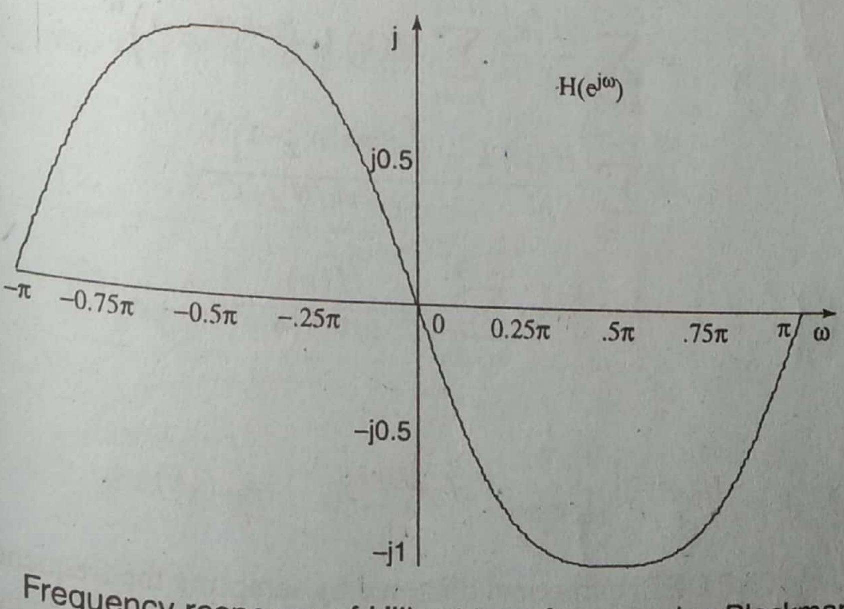

Fig. 6.55 Frequency response of Hilbert transformer using Blackman window

6.11.3 *Lattice structure of an FIR filter*

Let us consider an FIR filter with system function

$$H(z) = A_m(z) = 1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k} \quad m \geq 1$$

from which we have

$$Y(z) = X(z) \left[1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k} \right]$$

Taking inverse z-transform on both sides we get

$$y(n) = x(n) + \sum_{k=1}^{m} \alpha_m(k) x(n-k) \tag{6.168}$$

Interchange the role of input and output in Eq. (6.168), we obtain

$$x(n) = y(n) + \sum_{k=1}^{m} \alpha_m(k) y(n-k) \tag{6.169}$$

We find that the Eq. (6.169) describes an IIR system having the system function $H(z) = \frac{1}{A_m(z)}$, while the system described by the difference equation in Eq. (6.168) represents an FIR system with system function $H(z) = A_m(z)$.

Based on this, we use lattice structure system described in section (5.14.7) to obtain a lattice structure for an all-zero FIR system by interchanging the role of input and output.

For an all-pole filter the input

$$x(n) = f_N(n)$$

and the output

$$y(n) = f_0(n)$$

For an all-zero FIR system of order $M-1$ the input

$$x(n) = f_0(n)$$

and the output

$$y(n) = f_{M-1}(n)$$

For $m = 1$ the Eq. (6.168) reduces to

$$y(n) = x(n) + \alpha_1(1) x(n-1) \tag{6.170}$$

This output can also be obtained from a single-stage lattice filter shown in Fig. 6.79 from which we have

$$x(n) = f_0(n) = g_0(n)$$
$$y(n) = f_1(n) = f_0(n) + k_1 g_0(n-1)$$
$$= x(n) + k_1 x(n-1) \quad (6.171a)$$

and

$$g_1(n) = k_1 f_0(n) + g_0(n-1)$$
$$= k_1 x(n) + x(n-1) \quad (6.171b)$$

Comparing Eq. (6.170) with Eq. (6.171) we get

$$\alpha_1(0) = 1$$
$$\alpha_1(1) = k_1$$

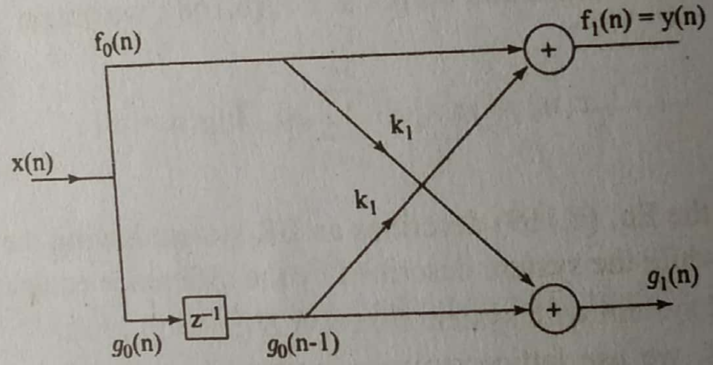

Fig. 6.79 Single stage all-zero lattice filter.

Now let us consider an FIR filter for which $m = 2$, then

$$y(n) = x(n) + \alpha_2(1)x(n-1) + \alpha_2(2)x(n-2) \quad (6.172)$$

By cascading two lattice stages as shown in Fig. 6.80 it is possible to obtain the output $y(n)$.

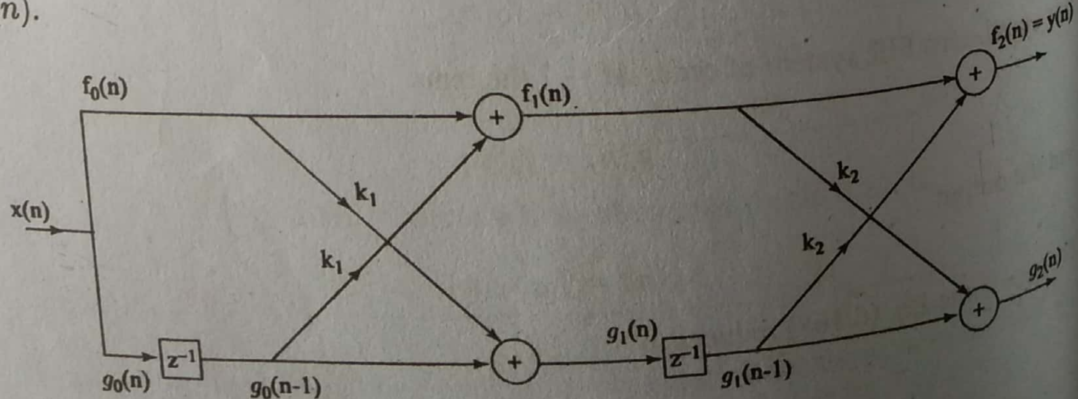

Fig. 6.80 Two stage all-zero lattice filter

From Fig. (6.80) the output from second stage is

$$y(n) = f_2(n) = f_1(n) + k_2 g_1(n-1) \tag{6.173}$$
$$g_2(n) = k_2 f_1(n) + g_1(n-1)$$

Substitute for $f_1(n)$ and $g_1(n-1)$ from Eq. (6.171a) and (6.171b) in Eq. (6.173) we get

$$y(n) = f_2(n) = x(n) + k_1 x(n-1) + k_2[k_1 x(n-1) + x(n-2)]$$
$$= x(n) + k_1(1 + k_2)x(n-1) + k_2 x(n-2) \tag{6.174}$$

Eq. (6.174) is identical to Eq. (6.172) from which we have

$$\alpha_2(0) = 1; \alpha_2(2) = k_2; \alpha_2(1) = k_1(1 + k_2)$$
$$= \alpha_1(1)[1 + \alpha_2(2)] \tag{6.175}$$

Similarly

$$g_2(n) = \alpha_2 x(n) + k_1(1 + k_2)x(n-1) + x(n-2) \tag{6.176}$$

From Eq. (6.174) and Eq. (6.176) we observe that the filter co-efficients for the lattice filter that produces $f_2(n)$ are $\{1, \alpha_2(1), \alpha_2(2)\}$ while the co-efficients for filter with output $g_2(n)$ are $\{\alpha_2(2), \alpha_2(1), 1\}$. We note that these two sets of co-efficients are in reverse order.

For a $M-1$ stage filter

$$f_0(n) = g_0(n)$$
$$f_m(n) = f_{m-1}(n) + k_m g_{m-1}(n-1) \qquad m = 1, 2, \ldots M-1$$
$$g_m(n) = k_m f_{m-1}(n) + g_{m-1}(n-1) \qquad m = 1, 2, \ldots M-1$$

The output of $M-1$ stage filter $y(n) = f_{M-1}(n)$ (6.177)

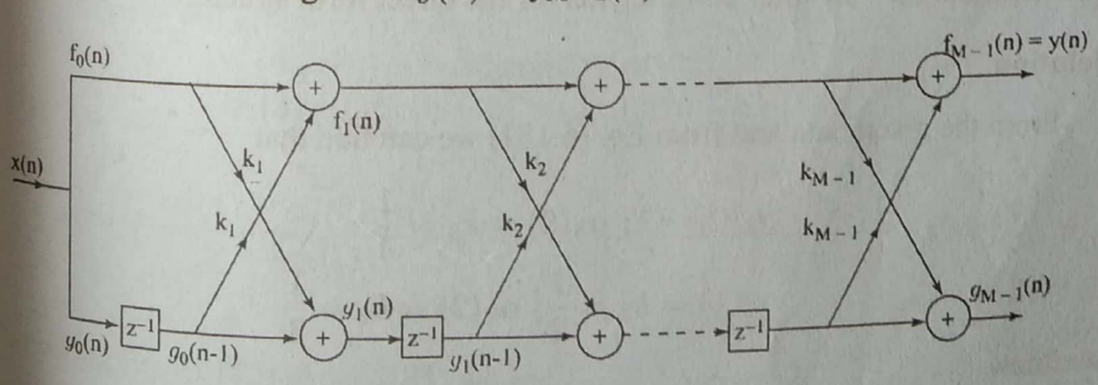

Fig. 6.81 All-zero lattice filter

Conversion of lattice co-efficients to direct form filter co-efficients

For $m = 3$ the Eq. (6.168) can be written as

$$y(n) = x(n) + \alpha_3(1)x(n-1) + \alpha_3(2)x(n-2) + \alpha_3(3)x(n-3) \tag{6.178}$$

We have

$$\begin{aligned}
y(n) = f_3(n) &= f_2(n) + k_3 g_2(n-1) \\
&= x(n) + \alpha_2(1)x(n-1) + \alpha_2(2)x(n-2) \\
&\quad + k_3\alpha_2(2)x(n-1) + k_3\alpha_2(1)x(n-2) + k_3 x(n-3) \\
&= x(n) + [\alpha_2(1) + k_3\alpha_2(1)]x(n-1) \\
&\quad + [\alpha_2(2) + k_3\alpha_2(1)]x(n-2) + k_3 x(n-3)
\end{aligned} \tag{6.179}$$

Comparing Eq. (6.178) and Eq. (6.179) we get

$$\begin{aligned}
\alpha_3(0) &= 1 \\
\alpha_3(1) &= \alpha_2(1) + k_3\alpha_2(2) \\
&= \alpha_2(1) + \alpha_3(3)\alpha_2(2) \\
\alpha_3(2) &= \alpha_2(2) + k_3\alpha_2(1) \\
&= \alpha_2(2) + \alpha_3(3)\alpha_2(1) \\
\alpha_3(3) &= k_3
\end{aligned} \tag{6.180}$$

From Eq. (6.175) and Eq. (6.180), for a general case we can find that

$$\begin{aligned}
\alpha_m(0) &= 1 \\
\alpha_m(m) &= k_m \\
\alpha_m(k) &= \alpha_{m-1}(k) + \alpha_m(m)\alpha_{m-1}(m-k)
\end{aligned} \tag{6.181}$$

The Eq. (6.181) can be used to convert the lattice filter co-efficients to direct form FIR filter co-efficients.

Example 6.28 Consider an FIR lattice filter with co-efficients $k_1 = \frac{1}{2}; k_2 = \frac{1}{3}; k_3 = \frac{1}{4}$. Determine the FIR filter co-efficients for the direct form structure.

Solution

From the given data and from Eq. (6.181) we can find that

$$\alpha_3(0) = 1;\ \alpha_3(3) = k_3 = \frac{1}{4}$$

$$\alpha_1(1) = k_1 = \frac{1}{2};\ \alpha_2(2) = k_2 = \frac{1}{3}$$

we know

$$\alpha_m(k) = \alpha_{m-1}(k) + k_m\alpha_{m-1}(m-k)$$

for $m = 2$ and $k = 1$

$$\begin{aligned}
\alpha_2(1) &= \alpha_1(1) + k_2\alpha_1(1) \\
&= \frac{1}{2} + \frac{1}{3}\cdot\frac{1}{2} = \frac{3+1}{6} = \frac{2}{3}
\end{aligned}$$

for $m = 3$ and $k = 1$

$$\alpha_3(1) = \alpha_2(1) + \alpha_3(3)\alpha_2(2)$$
$$= \frac{2}{3} + \frac{1}{4}\cdot\frac{1}{3} = \frac{8+1}{12} = \frac{3}{4}$$

for $m = 2$ and $k = 2$

$$\alpha_3(2) = \alpha_2(2) + \alpha_3(3)\alpha_2(1)$$
$$= \frac{1}{3} + \frac{1}{4}\cdot\frac{2}{3} = \frac{4+2}{12} = \frac{1}{2}$$

$\Rightarrow \alpha_3(0) = 1; \alpha_3(1) = \frac{3}{4}; \alpha_3(2) = \frac{1}{2}; \alpha_3(3) = \frac{1}{4}.$

Conversion of direct form FIR filter co-efficients to lattice co-efficients

For a 3-stage direct form FIR filter

$$y(n) = x(n) + \alpha_3(1)x(n-1) + \alpha_3(2)x(n-2) + \alpha_3(3)x(n-3) \qquad (6.182)$$

For a 3-stage lattice structure

$$y(n) = f_3(n) = f_2(n) + k_3 g_2(n-1)$$
$$g_3(n) = k_3 f_2(n) + g_2(n-1)$$

$$y(n) = f_3(n) = f_2(n) + k_3[g_3(n) - k_3 f_2(n)]$$
$$= f_2(n) = \frac{f_3(n) - k_3 g_3(n)}{1 - k_3^2}$$
$$= \frac{\begin{array}{c} x(n) + \alpha_3(1)x(n-1) + \alpha_3(2)x(n-2) + \alpha_3(3)x(n-3) \\ -k_3^2 x(n) - \alpha_3(3)\alpha_3(2)x(n-1) \\ -\alpha_3(3)\alpha_3(1)x(n-2) - \alpha_3(3)x(n-3) \end{array}}{1 - \alpha_3^2(3)}$$
$$= x(n) + \frac{\alpha_3(1) - \alpha_3(3)\alpha_3(2)}{1 - \alpha_3^2(3)}x(n-1) + \frac{\alpha_3(2) - \alpha_3(3)\alpha_3(1)}{1 - \alpha_3^2(3)}x(n-2) \qquad (6.183)$$

Comparing Eq. (6.172) and (6.183) we get

$$\alpha_2(0) = 1 \qquad (6.184)$$

$$\alpha_2(1) = \frac{\alpha_3(1) - \alpha_3(3)\alpha_3(2)}{1 - \alpha_3^2(3)} \qquad (6.185)$$

$$\alpha_2(2) = \frac{\alpha_3(2) - \alpha_3(3)\alpha_3(1)}{1 - \alpha_3^2(3)} \qquad (6.186)$$

In general for a m-stage filter

$$\alpha_{m-1}(0) = 1; k_m = \alpha_m(m)$$

$$\alpha_{m-1}(k) = \frac{\alpha_m(k) - \alpha_m(m)\alpha_m(m-k)}{1-\alpha_m^2(m)} 1 \le k \le m-1 \tag{6.187}$$

Example 6.29 An FIR filter is given by the difference equation

$$y(n) = 2x(n) + \frac{4}{5}x(n-1) + \frac{3}{2}x(n-2) + \frac{2}{3}x(n-3)$$

(AU ECE May'07)

Determine its lattice form.

Solution

Given

$$y(n) = 2x(n) + \frac{4}{5}x(n-1) + \frac{3}{2}x(n-2) + \frac{2}{3}x(n-3)$$

$$= 2\left[x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)\right]$$

$$= k_0\left[1 + \sum_{k=1}^{3}\alpha_m(k)x(n-k)\right]$$

where $k_0 = 2$

$$\alpha_3(0) = 1; \alpha_3(1) = \frac{2}{5}; \alpha_3(2) = \frac{3}{4}; \alpha_3(3) = \frac{1}{3}$$

From Eq. (6.187) we get

$$\alpha_2(0) = 1$$

$$k_3 = \alpha_2(3) = \frac{1}{3}$$

$$\alpha_{m-1}(k) = \frac{\alpha_m(k) - \alpha_m(m)\alpha_m(m-k}{1-\alpha_m^2(m)} \quad 1 \le k \le 2$$

For $m = 3$ and $k = 1$

$$\alpha_2(1) = \frac{\alpha_3(1) - \alpha_3(3)\alpha_3(2)}{1-\alpha_3^2(3)}$$

$$= \frac{\frac{2}{5} - \frac{1}{3}\cdot\frac{3}{4}}{1-\left(\frac{1}{3}\right)^2} = 0.16875$$

For $m = 3$ and $k = 2$

$$k_2 = \alpha_2(2) = \frac{\alpha_3(2) - \alpha_3(3)\alpha_3(1)}{1-\alpha_3^2(3)}$$

$$= \frac{\frac{3}{4} - \frac{1}{3}\cdot\frac{2}{5}}{1-\left(\frac{1}{3}\right)^2}$$

$$= \frac{111}{160} = 0.69375$$

For $m = 2$ and $k = 1$

$$k_1 = \alpha_1(1) = \frac{\alpha_2(1) - \alpha_2(2)\alpha_2(1)}{1 - \alpha_2^2(2)}$$

$$= \frac{\frac{27}{160} - \frac{111}{160} \cdot \frac{27}{160}}{1 - \left(\frac{111}{160}\right)^2} = 0.0996$$

Practice Problem 6.10 An FIR filter is given by the difference equation

$$y(n) = x(n) + \frac{1}{3}x(n-1) + \frac{1}{2}x(n-2) + \frac{1}{3}x(n-3)$$

Determine its lattice form.

6.11.4 *Polyphase realization of FIR filter*

Let us consider an FIR filter with impulse response having N coefficients. The system function of such a filter is given by

$$H(z) = \sum_{n=0}^{N-1} h(n)z^{-n} \tag{6.188}$$

If $N = 11$ then

$$H(z) = h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3} + h(4)z^{-4} + h(5)z^{-5}$$
$$+ h(6)z^{-6} + h(7)z^{-7} + h(8)z^{-8} + h(9)z^{-9} + h(10)z^{-10} \tag{6.189}$$

The above transfer function can be partitioned into two sub-signals, one sub signal containing even indexed coefficients and the other containing odd-indexed coefficients. That is

$$H(z) = h(0) + h(2)z^{-2} + h(4)z^{-4} + h(6)z^{-6} + h(8)z^{-8} + h(10)z^{-10}$$
$$+ h(1)z^{-1} + h(3)z^{-3} + h(5)z^{-5} + h(7)z^{-7} + h(9)z^{-9} \tag{6.190}$$
$$= h(0) + h(2)z^{-2} + h(4)z^{-4} + h(6)z^{-6} + h(8)z^{-8} + h(10)z^{-10}$$
$$+ z^{-1}[h(1) + h(3)z^{-2} + h(5)z^{-4} + h(7)z^{-6} + h(9)z^{-8}] \tag{6.191}$$
$$= P_0(z^2) + z^{-1}P_1(z^2) \tag{6.192}$$

where

$$P_0(z) = h(0) + h(2)z^{-1} + h(4)z^{-2} + h(6)z^{-3} + h(8)z^{-4} + h(10)z^{-5}$$
$$P_1(z) = h(1) + h(3)z^{-1} + h(5)z^{-2} + h(7)z^{-3} + h(9)z^{-4} \tag{6.193}$$